Assignment No	8
Title	A)Program to use callback function for Asynchronous file read B) Program using setTimeout to demonstrate callback function C)Program using SetInterval,ClearInterval,ClearTimeout to cancel execution of function
Objective	Learning and understanding.
Roll No	MCA2649

A)Program to use callback function for Asynchronous file read

Source Code:-

```
var fs=require("fs");

fs.writeFileSync('input.txt','writing a new file old will be overwritten');
fs.readFile('input.txt',function(ferr,data)
{
    if(ferr) console.error(ferr);
    console.log(data.toString());
    console.log("file reading done");
}
);
console.log("reading in program so doing something else");
var x = 5;
var y = 6;
console.log(`Sum of ${x} and ${y} is ${x+y}`);
console.log("Program ended");
```

output:-

```
PS D:\MCA2649\WT2649> node callbackread.js
Program ended
hello good morning
Node js is a javascript fro server side
React js is a open library for desiging front end
file reading done
```

```
PS D:\MCA2649\WT2649> node callbackread.js
reading in program so doing something else
Sum of 5 and 6 is 11
Program ended
writing a new file old will be overwritten
```

B) Program using setTimeout to demonstrate callback function

Source Code:-

```
var calculate=function(x,y,display){
    let sum=x+y;
    setTimeout(function(){display(sum)},20000);
    //will execute after 20 seconds
    console.log("VIP task taking place");
    //function call leaves this block
}
calculate(5,10,function(res){
    console.log("Sum of 5 and 10 is ",res);
    console.log("program ends");
});
//call comes here and prints the following
console.log("doing something else......");
console.log("Please wait.......");
```

output:-

```
PS D:\MCA2649\WT2649> node callback2.js
VIP task taking place
doing something else......
Please wait.......
Sum of 5 and 10 is  15
program ends
```

C)Program using SetInterval,ClearInterval,ClearTimeout to cancel execution of function

Source Code:-

```
let i=0;
var inter=setInterval(function()
{
    ++i;
    console.log("Hello There");
    if(i==10) clearInterval(inter);
},3000);
var t=setTimeout(function()
{
    console.log("Time out example!");
},2000);

clearTimeout(t);
console.log("Program is running!");
process.nextTick(function()
{
    console.log("Urgent task PM/CM");
});
console.log("Hello => no 1");
setImmediate(()=>{
    console.log("Running before the timeout => no 3");
});
setTimeout(()=>{
    console.log("The timeout running last => no 4");
},100);
process.nextTick(()=>{
    console.log("Running at next tick => no 2");
});
```

output:-

```
PS D:\MCA2649\WT2649> node callback3.js
Program is running!
Hello => no 1
Urgent task PM/CM
Running at next tick => no 2
Running before the timeout => no 3
The timeout running last => no 4
Hello There
Hello There
Hello There
Hello There
Hello There
Hello There
Hello There
Hello There
Hello There
Hello There
PS D:\MCA2649\WT2649>
```